Física Espacial — fórmulas

$\mu = GM, \quad h = L/m$

CANTIDAD DE MOVIMIENTO Y CHOQUES

$$\boldsymbol{p} = m\boldsymbol{v}, \quad \sum \boldsymbol{F} = \frac{d\boldsymbol{p}}{dt}$$

$$\boldsymbol{J} = \int \boldsymbol{F}dt = \Delta\boldsymbol{p} = \boldsymbol{F}_{\text{med}}\Delta t$$

$$\boldsymbol{r}_{\text{cm}} = \frac{\sum m_i \boldsymbol{r}_i}{M}, \quad \boldsymbol{P} = M\boldsymbol{v}_{\text{cm}}$$

$$\sum \boldsymbol{F}_{\text{ext}} = M\boldsymbol{a}_{\text{cm}}, \quad \sum \boldsymbol{F}_{\text{ext}} = 0 \Rightarrow \boldsymbol{P} = \text{cte}$$

$$m_1\boldsymbol{v}_1 + m_2\boldsymbol{v}_2 = m_1\boldsymbol{v}_1' + m_2\boldsymbol{v}_2'$$

$$\boldsymbol{v}_1' = \boldsymbol{v}_2' = \frac{\boldsymbol{P}}{M}, \quad K_{\text{perd}} = \frac{1}{2}m_r v_{\text{rel}}^2$$

$$m_r = \frac{m_1 m_2}{m_1 + m_2}, \quad \boldsymbol{v}_{\text{rel}} = \boldsymbol{v}_A - \boldsymbol{v}_B$$

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + K_{\text{rel}}, \quad K = \frac{p^2}{2m}$$

COHETE

$$\mu = -\frac{dM}{dt}, \quad f = \mu|v_r| = I_{\text{sp}} g_0 \mu, \quad I_{\text{sp}} = \frac{|v_r|}{g_0}$$

$$M\frac{dV}{dt} = \mu|v_r| - Mg, \quad M(t^*) = \frac{\mu|v_r|}{g}$$

$$M(t) = M_0 - \mu t, \quad a(t) = \frac{\mu|v_r|}{M_0 - \mu t} - g$$

$$V(t) = V_0 + |v_r| \ln\frac{M_0}{M(t)} - gt$$

$$y(t) = y_0 + V_0 t - \frac{1}{2}gt^2 + |v_r|\left[t - \frac{M(t)}{\mu}\ln\frac{M_0}{M(t)}\right]$$

$$\Delta V = |v_r| \ln\frac{M_0}{M_f}, \quad \frac{M_0}{M_f} = \exp\left(\frac{\Delta V}{|v_r|}\right)$$

$$\Delta V_{\text{tot}} = \sum \Delta V_i, \quad \boldsymbol{F}_{\text{ext}} = \frac{d(M\boldsymbol{v})}{dt}$$

TRABAJO Y ENERGÍA

$$W = \int \boldsymbol{F}\cdot d\boldsymbol{l}, \quad W_{\text{tot}} = \Delta K, \quad P = \boldsymbol{F}\cdot\boldsymbol{v}$$

$$\boldsymbol{F} = -\nabla U, \quad E = K + U, \quad W_{\text{no cons}} = \Delta E$$

MOMENTO ANGULAR

$$\boldsymbol{L}_O = \boldsymbol{r}\times m\boldsymbol{v}, \quad L = mvr\sin\varphi = mvd$$

$$\sum \boldsymbol{\tau}_O = \frac{d\boldsymbol{L}_O}{dt}, \quad \int \boldsymbol{\tau}dt = \Delta\boldsymbol{L}$$

$$h = rv_\perp = rv\cos\gamma = r^2\dot\theta = \text{cte}, \quad \frac{dA}{dt} = \frac{h}{2}$$

$$r_P v_P = r_A v_A$$

ROTACIÓN Y GIRÓSCOPO

$$v = \omega r, \quad a_t = \alpha r, \quad a_{\text{rad}} = \omega^2 r, \quad K = \frac{1}{2}I\omega^2$$

$$I = \sum m_i r_i^2, \quad I_P = I_{\text{cm}} + Md^2$$

varilla (cm)	$ML^2/12$	aro	MR^2
varilla (ext)	$ML^2/3$	disco	$MR^2/2$
esfera	$2MR^2/5$	esf. hueca	$2MR^2/3$

$$\tau = rF\sin\varphi, \quad \sum \tau_z = I\alpha_z, \quad L = I\omega$$

$$\sum \boldsymbol{\tau}_{\text{ext}} = \frac{d\boldsymbol{L}}{dt}$$

$$\sum \boldsymbol{\tau}_{\text{ext}} = 0 \Rightarrow \sum I_i\omega_i = \text{cte}$$

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2, \quad v_{\text{cm}} = R\omega$$

$$\Delta\boldsymbol{L} = \boldsymbol{\tau}\Delta t, \quad \Omega = \frac{\tau}{L} = \frac{Mgr}{I\omega}$$

GRAVITACIÓN

$$\boldsymbol{F}_g = -\frac{GMm}{r^2}\hat{\boldsymbol{r}}, \quad U = -\frac{GMm}{r}$$

$$g_0 = \frac{GM}{R^2} = \frac{\mu}{R^2}, \quad g(h) = g_0\left(\frac{R}{R+h}\right)^2$$

$$v_c = \sqrt{\frac{\mu}{r}}, \quad v_{\text{esc}} = \sqrt{\frac{2\mu}{r}} = \sqrt{2}\,v_c, \quad T = \frac{2\pi r}{v_c}$$

$$\Delta E_{R\to r} = \mu m\left(\frac{1}{R} - \frac{1}{2r}\right)$$

$$\Delta E_{r_1\to r_2} = \frac{\mu m}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

DOS CUERPOS

$$\ddot{\boldsymbol{r}} = -\frac{\mu}{r^3}\boldsymbol{r}, \quad \mu = G(m_1 + m_2) \approx Gm_1$$

$$\boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2, \quad \boldsymbol{r}_1 = \frac{m_2}{M}\boldsymbol{r}, \quad \boldsymbol{r}_2 = -\frac{m_1}{M}\boldsymbol{r}$$

$$E = \frac{1}{2}m\dot r^2 + \frac{L^2}{2mr^2} + U(r), \quad U_{\text{ef}} = U + \frac{L^2}{2mr^2}$$

ÓRBITAS

$$r = \frac{p}{1 + e\cos\nu}, \quad p = \frac{h^2}{\mu} = a(1-e^2)$$

$$\cos\nu = \frac{1}{e}\left(\frac{p}{r} - 1\right), \quad e\sin\nu = \frac{hv_r}{\mu}$$

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}, \quad v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$$

$$e = \sqrt{1 + \frac{2\varepsilon h^2}{\mu^2}} = \frac{r_A - r_P}{r_A + r_P} = \frac{r_P v_P^2}{\mu} - 1$$

$$e^2 = \left(\frac{p}{r} - 1\right)^2 + \left(\frac{hv_r}{\mu}\right)^2$$

$$e = \frac{r_2 - r_1}{r_1\cos\nu_1 - r_2\cos\nu_2}$$

$$h = rv\cos\gamma = r_P v_P = \sqrt{\mu p} = \sqrt{\mu a(1-e^2)}$$

$$r_P = a(1-e) = \frac{p}{1+e}, \quad r_A = a(1+e) = \frac{p}{1-e}$$

$$a = \frac{r_P + r_A}{2}, \quad b = a\sqrt{1-e^2} = \sqrt{r_P r_A}$$

$$\frac{1}{2}v_P^2 - \frac{\mu}{r_P} = \frac{1}{2}v_A^2 - \frac{\mu}{r_A}$$

$$v_P = \sqrt{\frac{2\mu r_A}{r_P(r_P + r_A)}}, \quad v_A = \frac{r_P}{r_A}v_P$$

$$v_P = \sqrt{\frac{\mu}{a}\frac{1+e}{1-e}}, \quad v_A = \sqrt{\frac{\mu}{a}\frac{1-e}{1+e}}$$

$$v_\perp = \frac{h}{r} = \frac{\mu}{h}(1 + e\cos\nu), \quad v_r = \frac{\mu}{h}e\sin\nu$$

$$\tan\gamma = \frac{v_r}{v_\perp} = \frac{e\sin\nu}{1 + e\cos\nu}$$

HIPÉRBOLA Y ESCAPE

$$e = 1: \quad \varepsilon = 0, \quad v = v_{\text{esc}} = \sqrt{\frac{2\mu}{r}}$$

$$\varepsilon = \frac{v_\infty^2}{2} = \frac{\mu}{2a}, \quad v^2 = v_{\text{esc}}^2 + v_\infty^2, \quad C_3 = v_\infty^2$$

$$a = \frac{p}{e^2 - 1} = \frac{\mu}{v_\infty^2}, \quad r_P = a(e-1)$$

$$e = 1 + \frac{r_P v_\infty^2}{\mu}$$

$$\nu_\infty = \arccos\left(-\frac{1}{e}\right), \quad \delta = 2\arcsin\left(\frac{1}{e}\right)$$

$$\Delta v_{\text{esc}} = \sqrt{\frac{2\mu}{r}} - v$$

KEPLER

$$T = 2\pi\sqrt{\frac{a^3}{\mu}} = \frac{2\pi ab}{h}, \quad n = \frac{2\pi}{T} = \sqrt{\frac{\mu}{a^3}}$$

$$\tan\frac{E}{2} = \sqrt{\frac{1-e}{1+e}}\tan\frac{\nu}{2}, \quad \cos E = \frac{e + \cos\nu}{1 + e\cos\nu}$$

$$M_e = E - e\sin E = n(t - t_P)$$

$$r = a(1 - e\cos E)$$

$$T_{\text{sid}} = 24\text{ h} \cdot \frac{360^\circ}{360{,}986^\circ} = 23{,}934\text{ h}$$

MANIOBRAS

$$v_P = \left[2\mu\frac{r_A}{r_P}\cdot\frac{1}{r_A + r_P}\right]^{1/2}, \quad v_A = \frac{r_P}{r_A}v_P$$

$$v_{cP} = \sqrt{\frac{\mu}{r_P}}, \quad v_{cA} = \sqrt{\frac{\mu}{r_A}}$$

$$\Delta v_1 = v_P - v_{cP}, \quad \Delta v_2 = v_{cA} - v_A$$

$$v_P > v_{cP} \Leftrightarrow \frac{r_A}{r_A + r_P} > \frac{1}{2}$$

$$a_t = \frac{r_P + r_A}{2}, \quad t_v = \frac{T_t}{2} = \pi\sqrt{\frac{a_t^3}{\mu}}$$

$$\varphi = 180^\circ - n_A t_v, \quad n_A = 360^\circ/T_A$$

$$\frac{T'}{T} = 1 - \frac{\Delta\varphi}{360^\circ}, \quad a' = r\left(\frac{T'}{T}\right)^{2/3}$$

$$r_{P'} = 2a' - r$$

ELEMENTOS ORBITALES

$$(\boldsymbol{r},\boldsymbol{v}) \leftrightarrow (a, e, i, \Omega, \omega, \nu_0)$$

$$\boldsymbol{A}\times\boldsymbol{B} = \begin{vmatrix} \hat I & \hat J & \hat K \\ A_I & A_J & A_K \\ B_I & B_J & B_K \end{vmatrix}$$

$$= (A_J B_K - A_K B_J)\hat I - (A_I B_K - A_K B_I)\hat J + (A_I B_J - A_J B_I)\hat K$$

$$|\boldsymbol{A}\times\boldsymbol{B}| = AB\sin\alpha, \quad \boldsymbol{A}\times\boldsymbol{B} = -\boldsymbol{B}\times\boldsymbol{A}$$

$$\hat I\times\hat J = \hat K, \quad \hat J\times\hat K = \hat I, \quad \hat K\times\hat I = \hat J$$

$$\boldsymbol{A}\cdot\boldsymbol{B} = A_I B_I + A_J B_J + A_K B_K = AB\cos\alpha$$

$$\boldsymbol{r}\cdot\boldsymbol{v} = rv_r = rv\sin\gamma$$

$$\boldsymbol{h} = \boldsymbol{r}\times\boldsymbol{v}, \quad \boldsymbol{n} = \hat K\times\boldsymbol{h} = (-h_J, h_I, 0)$$

$$\boldsymbol{e} = \frac{1}{\mu}\left[\left(v^2 - \frac{\mu}{r}\right)\boldsymbol{r} - (\boldsymbol{r}\cdot\boldsymbol{v})\boldsymbol{v}\right]$$

$$p = \frac{h^2}{\mu}, \quad e = |\boldsymbol{e}|, \quad a = \frac{p}{1-e^2} = -\frac{\mu}{2\varepsilon}$$

$$\cos i = \frac{h_K}{h}$$

$$\cos\Omega = \frac{n_I}{n} \qquad (n_J < 0 \Rightarrow \Omega > 180^\circ)$$

$$\cos\omega = \frac{\boldsymbol{n}\cdot\boldsymbol{e}}{ne} \qquad (e_K < 0 \Rightarrow \omega > 180^\circ)$$

$$\cos\nu_0 = \frac{\boldsymbol{e}\cdot\boldsymbol{r}}{er} \qquad (\boldsymbol{r}\cdot\boldsymbol{v} < 0 \Rightarrow \nu_0 > 180^\circ)$$

$$\Pi = \Omega + \omega \qquad (i = 0)$$

$$u_0 = \omega + \nu_0 \qquad (e = 0)$$

$$l_0 = \Omega + \omega + \nu_0 \qquad (i = e = 0)$$

$$\cos\Pi = \frac{e_I}{e} \qquad (e_J < 0 \Rightarrow \Pi > 180^\circ)$$

$$\cos u_0 = \frac{\boldsymbol{n}\cdot\boldsymbol{r}}{nr} \qquad (r_K < 0 \Rightarrow u_0 > 180^\circ)$$

$$\cos l_0 = \frac{r_I}{r} \qquad (r_J < 0 \Rightarrow l_0 > 180^\circ)$$

$$\boldsymbol{r} = r\cos\nu\,\hat P + r\sin\nu\,\hat Q$$

$$\boldsymbol{v} = \frac{\mu}{h}\left[-\sin\nu\,\hat P + (e + \cos\nu)\hat Q\right]$$

CONSTANTES

G	$6{,}674\times10^{-11}$ N·m²/kg²
μ_T, R_T	398 600 km³/s², 6378 km
g_0	9,80665 m/s²
μ_{Luna}	4903 km³/s², $d_{\text{T–L}} \approx 384\,400$ km
μ_{Sol}	$1{,}327\times10^{11}$ km³/s²
1 UA	$149{,}6\times10^{6}$ km

MARCOS NO INERCIALES

$$\boldsymbol{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$
$$\boldsymbol{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\hat{r} + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\hat{\theta}$$
$$m\boldsymbol{a}' = \boldsymbol{F} - m\boldsymbol{A}$$
$$m\boldsymbol{a}_{\text{rel}} = \boldsymbol{F} - 2m\boldsymbol{\Omega}\times\boldsymbol{v}_{\text{rel}} - m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) - m\dot{\boldsymbol{\Omega}}\times\boldsymbol{r}$$

ESFERA DE INFLUENCIA Y CÓNICAS PARCHEADAS

$$r_{\text{SOI}} = R\left(\frac{m_p}{m_s}\right)^{2/5}, \quad r_{\text{SOI,T}} \approx 925\,000 \text{ km}$$
$$\boldsymbol{v}_\infty = \boldsymbol{V}_{\text{nave}} - \boldsymbol{V}_{\text{planeta}}$$
$$v_P = \sqrt{v_\infty^2 + \frac{2\mu}{r_P}}, \quad v_c = \sqrt{\mu/r_P}$$
$$e = 1 + \frac{r_P v_\infty^2}{\mu}$$
$$\Delta v = v_P - v_c = v_c\left(\sqrt{2 + (v_\infty/v_c)^2} - 1\right)$$

COEFICIENTES DE LAGRANGE

$$\boldsymbol{r} = f\boldsymbol{r}_0 + g\boldsymbol{v}_0, \quad \boldsymbol{v} = \dot{f}\boldsymbol{r}_0 + \dot{g}\boldsymbol{v}_0$$
$$f\dot{g} - \dot{f}g = 1$$
$$f = 1 - \frac{\mu r}{h^2}(1 - \cos\Delta\nu), \quad g = \frac{rr_0}{h}\sin\Delta\nu$$
$$\dot{g} = 1 - \frac{\mu r_0}{h^2}(1 - \cos\Delta\nu)$$

TRES CUERPOS RESTRINGIDO

$$\Omega = \sqrt{\mu/r_{12}^3}, \quad \pi_i = \frac{m_i}{m_1 + m_2}$$
$$x_1 = -\pi_2 r_{12}, \quad x_2 = \pi_1 r_{12}$$
$$\ddot{x} - 2\Omega\dot{y} - \Omega^2 x = -\frac{\mu_1}{r_1^3}(x + \pi_2 r_{12}) - \frac{\mu_2}{r_2^3}(x - \pi_1 r_{12})$$
$$\ddot{y} + 2\Omega\dot{x} - \Omega^2 y = -\left(\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3}\right)y$$
$$L_{4,5}: \left(\frac{r_{12}}{2} - \pi_2 r_{12}, \pm\frac{\sqrt{3}}{2}r_{12}\right)$$
$$r_{\text{Hill}} = r_{12}\left(\frac{m_2}{3m_1}\right)^{1/3}, \quad \frac{m_1}{m_2} + \frac{m_2}{m_1} \geq 25$$
$$C = \frac{v^2}{2} - \frac{\Omega^2(x^2+y^2)}{2} - \frac{\mu_1}{r_1} - \frac{\mu_2}{r_2}$$

CUERPO RÍGIDO EN 3D

$$\boldsymbol{v}_B = \boldsymbol{v}_A + \boldsymbol{\omega}\times\boldsymbol{r}_{B/A}$$
$$\boldsymbol{a}_B = \boldsymbol{a}_A + \boldsymbol{\alpha}\times\boldsymbol{r}_{B/A} + \boldsymbol{\omega}\times\left(\boldsymbol{\omega}\times\boldsymbol{r}_{B/A}\right)$$
$$\left(\dot{\boldsymbol{Q}}\right)_{\text{fijo}} = \left(\dot{\boldsymbol{Q}}\right)_{\text{rot}} + \boldsymbol{\Omega}\times\boldsymbol{Q}$$
$$\begin{pmatrix} H_x \\ H_y \\ H_z \end{pmatrix} = \begin{pmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{xy} & I_y & -I_{yz} \\ -I_{xz} & -I_{yz} & I_z \end{pmatrix}\begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$
$$\boldsymbol{H}_O = \bar{\boldsymbol{r}}\times m\bar{\boldsymbol{v}} + \boldsymbol{H}_G, \quad T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\boldsymbol{\omega}\cdot\boldsymbol{H}_G$$
$$\sum M_x = I_x\dot{\omega}_x - (I_y - I_z)\omega_y\omega_z$$
$$\sum M_y = I_y\dot{\omega}_y - (I_z - I_x)\omega_z\omega_x$$
$$\sum M_z = I_z\dot{\omega}_z - (I_x - I_y)\omega_x\omega_y$$
$$\sum \boldsymbol{M}_O = \dot{\varphi}\sin\theta\left[I'\dot{\psi} + (I' - I)\dot{\varphi}\cos\theta\right]\hat{f}$$
$$\theta = 90°: \quad \sum \boldsymbol{M}_O = I'\dot{\varphi}\dot{\psi}\hat{f}$$
$$\dot{\varphi} = H/I, \quad \tan\gamma = \frac{I}{I'}\tan\theta$$